Ziad Mohamed Sheashaa

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Summary

AI Engineer specializing in Agentic & Generative AI, LLM orchestration (LangGraph, Multi-Agent architectures), advanced RAG systems, and production API design. Combines state-of-the-art LLM engineering with a solid foundation in computer vision, edge AI deployment (ONNX, YOLO), and high-performance backend systems (FastAPI, Python, C++).

Education

Suez Canal University — Ismailia, Egypt
Bachelor of Engineering in Computer Engineering — *Sep 2021 - Jul 2026*

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, HTML/CSS, Bash
Agentic & Generative AI: LangGraph, LangChain, Multi-Agent Systems, RAG, Model Context Protocol (MCP), Vector DBs (ChromaDB, FAISS), Prompt Engineering, Structured Outputs (Pydantic/Zod), Tool Calling
AI/ML & Computer Vision: PyTorch, TensorFlow, YOLOv8/YOLO, OpenCV, ONNX Runtime, Scikit-learn, Vision-Language Models (VLMs), Hugging Face
Backends & Frameworks: FastAPI, REST APIs, WebSockets, Next.js, React, ROS2
Tools & Infrastructure: Git & GitHub, Linux, Docker, MongoDB, SQLite, CMake, Postman
Core Engineering: Data Structures & Algorithms, Object-Oriented Programming (OOP), SOLID Principles, System Architecture, Design Patterns

Experience

AI Engineer — Mar 2026 - Present
Nexus — *Ismailia, Egypt*

- Architected and deployed production-grade multi-agent systems and stateful workflows using LangGraph and LangChain, orchestrating frontier LLMs (OpenAI, Anthropic, Gemini) with robust error recovery and dynamic tool routing.
- Engineered high-throughput FastAPI backends serving enterprise RAG pipelines, incorporating hybrid search (vector embeddings + BM25 keyword retrieval) and cross-encoder reranking to minimize hallucinations.
- Implemented Model Context Protocol (MCP) servers and structured output parsing pipelines (Pydantic), guaranteeing type-safe execution between LLM reasoning layers and downstream software services.
- Optimized inference latency and token consumption through prompt caching, dynamic context pruning, and automated evaluation benchmarks.

Generative AI Trainee — Jul 2025 - Aug 2025
Information Technology Institute (ITI) — *Egypt*

- Completed intensive, specialized training on transformer architectures, LLM fine-tuning methodologies (LoRA/QLoRA), and advanced prompt engineering.
- Engineered end-to-end RAG applications leveraging vector databases, semantic chunking strategies, and automated evaluation metrics to assess retrieval fidelity.

Artificial Intelligence Trainee — Jul 2024 - Aug 2024
Information Technology Institute (ITI) — *Egypt*

- Completed intensive foundational AI training spanning statistical machine learning, deep neural networks (CNNs, RNNs), and computer vision pipelines.
- Trained and evaluated computer vision and supervised learning models using PyTorch and Scikit-learn, executing rigorous data preprocessing and validation loops.

Projects

Nexcut *(Agentic Desktop Video Editor)* | *Gemma, ONNX Runtime, MCP, C++20, Electron* Jul 2026 - Present

- Architected the AI agent subsystem for an on-device video editor, translating natural-language instructions into multi-track timeline edits via MCP-compliant tool executions.
- Designed a deterministic object-grounding protocol (UUID-to-short-ref resolution) paired with a Zod validation-feedback retry loop, completely mitigating tool-call hallucinations on quantized local models.
- Engineered high-speed local inference pipelines utilizing ONNX Runtime for lightweight models (Gemma, Qwen) to deliver sub-second offline editing commands.

Vision-Based Desktop Automation | *Gemini 2.0 Flash, GUI-Actor-3B (Qwen2.5-VL), PyTorch* May 2026

- Built a hierarchical GUI automation agent utilizing Gemini 2.0 Flash as a strategic planner and GUI-Actor-3B as a real-time, on-screen visual grounding model.
- Developed an autonomous execution engine translating UI bounding coordinates into precise OS-level mouse and keyboard actions with visual self-reflection loops to verify state transitions.

SafeSpace *(Graduation Project)* | *YOLO, Raspberry Pi 5, FastAPI, OpenCV, Python* Sep 2025 - Jul 2026

- Architected an edge-computing security system deployed on Raspberry Pi 5, processing multi-camera real-time streams for automated surveillance and access control.
- Engineered an automated license-plate recognition (ALPR) pipeline using YOLO object detection and OCR, backed by an asynchronous FastAPI REST service with sub-second logging.

TERRA Climate Education Platform | *FastAPI, Scikit-learn, Gemini API, Next.js* Oct 2025

- Developed an educational platform analyzing NASA climate data with Scikit-learn predictive models to forecast environmental metrics.
- Integrated an interactive Gemini API chatbot with system prompt persona design alongside an interactive simulation that visualizes real-time climate impact metrics.

Real-Time Facial Detection & Analysis | *C++, Qt 6, OpenCV, CMake* Dec 2024

- Built a real-time facial feature detection pipeline in modern C++ with OpenCV, optimizing multi-scale image processing and frame-buffer throughput.
- Engineered a responsive Qt desktop GUI providing live webcam feeds, bounding overlays, and real-time inference latency diagnostics.

Volunteering

Autonomous Driving Team Leader Oct 2024 - Present

SCU Racing Team · Suez Canal University *Ismailia, Egypt*

- Lead a 20-engineer division developing autonomous racing software stacks using ROS2, computer vision, and sensor fusion.
- Define software architectures, direct perception/planning/control modules, and conduct peer code reviews to ensure system reliability.

AI Instructor Sep 2025 - Present

IEEE - SCU Student Branch *Ismailia, Egypt*

- Designed and instructed comprehensive technical workshops on machine learning and computer vision fundamentals to 50+ students.
- Mentored undergraduate students through practical project implementations, model debugging, and algorithmic problem solving.

Autonomous Driving Team Member Mar 2024 - Oct 2024

SCU Racing Team · Suez Canal University *Ismailia, Egypt*

- Developed core perception nodes and closed-loop control algorithms in ROS2 and Python, contributing to a 5th place award in the EVER Gen 5th international autonomous electric vehicle competition.

AI Technical Member Oct 2023 - Sep 2025

IEEE - SCU Student Branch *Ismailia, Egypt*

- Developed a full-stack movie recommendation engine using unsupervised ML (Python, Scikit-learn, MongoDB) and delivered introductory AI training sessions.

Certifications & Coursework

Certifications: Google AI Professional (Coursera, Jul 2026), HCIA-AI (Huawei, Jan 2023), Career Essentials in Generative AI (Microsoft & LinkedIn, Nov 2023)

Coursework: ROS2 & Software Engineering (Robotics Corner), Deep Learning & Machine Learning (E.Camp), Artificial Intelligence (Zewail City of Science and Technology, Huawei)

Languages

Arabic: Native **English**: Advanced / Professional Working Proficiency